

SPENVIS 4.6.14.3582 10-May-2026 09:56:27

Solar cell damage equivalent fluences

Project: DEMO1

iss_0.5sys_duration

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Mission segment 1

Trapped protons for [P_{max}](#) [V_{oc}](#) [I_{sc}](#)[Trapped electrons](#)Solar protons for [P_{max}](#) [V_{oc}](#) [I_{sc}](#)

Cell: Si
Electron/proton damage ratios: $P_{\max} = 3000.0$ $V_{oc} = 3000.0$ $I_{sc} = 3000.0$
Coverglass
material: fused silica density: 2.20 (g/cm ³)

Summary of 1 MeV equivalent electron fluences (cm⁻²)

Coverglass thickness			Total			Trapped electrons	Trapped protons			Solar protons		
g cm ⁻²	mils	micron	P _{max}	V _{oc}	I _{sc}	P _{max} , V _{oc} , I _{sc}	P _{max}	V _{oc}	I _{sc}	P _{max}	V _{oc}	I _{sc}
0.0000	0.0	0.00	8.809E+12	8.809E+12	9.246E+11	3.562E+10	8.773E+12	8.773E+12	8.890E+11	1.000E-12	1.000E-12	1.000E-12
0.0056	1.0	25.40	5.985E+11	5.985E+11	3.009E+11	3.076E+10	5.677E+11	5.677E+11	2.702E+11	1.000E-12	1.000E-12	1.000E-12
0.0168	3.0	76.20	3.148E+11	3.148E+11	2.034E+11	2.573E+10	2.890E+11	2.890E+11	1.777E+11	1.000E-12	1.000E-12	1.000E-12
0.0335	6.0	152.40	2.239E+11	2.239E+11	1.639E+11	2.122E+10	2.027E+11	2.027E+11	1.427E+11	1.000E-12	1.000E-12	1.000E-12
0.0671	12.0	304.80	1.666E+11	1.666E+11	1.345E+11	1.599E+10	1.506E+11	1.506E+11	1.185E+11	1.000E-12	1.000E-12	1.000E-12
0.1118	20.0	508.00	1.320E+11	1.320E+11	1.148E+11	1.201E+10	1.200E+11	1.200E+11	1.028E+11	1.000E-12	1.000E-12	1.000E-12
0.1676	30.0	762.00	1.143E+11	1.143E+11	1.036E+11	8.972E+09	1.053E+11	1.053E+11	9.461E+10	1.000E-12	1.000E-12	1.000E-12
0.3353	60.0	1524.00	9.300E+10	9.300E+10	8.703E+10	4.384E+09	8.862E+10	8.862E+10	8.265E+10	1.000E-12	1.000E-12	1.000E-12
User defined coverglass thickness												
0.0220	3.9	100.00	2.754E+11	2.754E+11	1.869E+11	2.386E+10	2.515E+11	2.515E+11	1.631E+11	0.000E+00	0.000E+00	0.000E+00

[Trapped protons](#) for [mission segment 1](#) for P_{max} (integration restricted to the input energy range). D(E,...) values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

			0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micron	
Energy Int. (MeV)	Int. Fluence (cm ⁻²)	Diff. Fluence (cm ⁻²)	D(E, 0.0)	Eq. Fluence (cm ⁻²)	D(E, 1.0)	Eq. Fluence (cm ⁻²)	D(E, 3.0)	Eq. Fluence (cm ⁻²)	D(E, 6.0)	Eq. Fluence (cm ⁻²)	D(E, 12.0)	Eq. Fluence (cm ⁻²)	D(E, 20.0)	Eq. Fluence (cm ⁻²)	D(E, 30.0)	Eq. Fluence (cm ⁻²)	D(E, 60.0)	Eq. Flu (cm)
0.10	0.20	3.371E+09	1.329E+09	6.17E-01	8.203E+08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.20	0.30	2.042E+09	6.734E+08	7.82E-01	5.269E+08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.30	0.40	1.369E+09	4.101E+08	9.25E-01	3.794E+08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.40	0.60	9.585E+08	3.831E+08	1.12E+00	4.297E+08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.60	0.80	5.754E+08	1.521E+08	1.40E+00	2.123E+08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
0.80	1.00	4.233E+08	8.545E+07	1.66E+00	1.421E+08	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.00	1.20	3.378E+08	3.728E+07	1.89E+00	7.044E+07	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.20	1.30	3.006E+08	1.504E+07	2.04E+00	3.064E+07	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
1.30	1.40	2.855E+08	1.325E+07	2.12E+00	2.810E+07	9.77E-02	1.295E+06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

1.40	1.60	2.723E+08	2.065E+07	2.23E+00	4.594E+07	3.41E-01	7.032E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.60	1.80	2.516E+08	1.540E+07	2.35E+00	3.627E+07	7.63E-01	1.176E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.80	2.00	2.362E+08	1.298E+07	2.46E+00	3.188E+07	1.15E+00	1.489E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.00	2.20	2.232E+08	6.172E+06	2.54E+00	1.565E+07	1.47E+00	9.096E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.20	2.40	2.171E+08	5.486E+06	2.59E+00	1.422E+07	1.74E+00	9.538E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.40	2.60	2.116E+08	4.924E+06	2.63E+00	1.295E+07	1.95E+00	9.609E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.60	2.80	2.067E+08	4.457E+06	2.65E+00	1.181E+07	2.12E+00	9.438E+06	8.86E-02	3.948E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.80	3.00	2.022E+08	4.063E+06	2.65E+00	1.076E+07	2.24E+00	9.116E+06	4.16E-01	1.689E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.00	3.20	1.981E+08	3.296E+06	2.62E+00	8.631E+06	2.34E+00	7.701E+06	7.60E-01	2.505E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.20	3.40	1.948E+08	3.046E+06	2.56E+00	7.803E+06	2.40E+00	7.298E+06	1.07E+00	3.273E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.40	3.60	1.918E+08	2.829E+06	2.48E+00	7.003E+06	2.42E+00	6.840E+06	1.34E+00	3.798E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.60	3.80	1.890E+08	2.637E+06	2.36E+00	6.233E+06	2.40E+00	6.340E+06	1.56E+00	4.123E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.80	4.00	1.863E+08	2.468E+06	2.23E+00	5.503E+06	2.35E+00	5.809E+06	1.74E+00	4.293E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.00	4.20	1.839E+08	1.659E+06	2.09E+00	3.468E+06	2.27E+00	3.764E+06	1.87E+00	3.109E+06	1.28E-01	2.121E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.20	4.40	1.822E+08	1.568E+06	1.96E+00	3.067E+06	2.16E+00	3.379E+06	1.96E+00	3.080E+06	4.10E-01	6.426E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.40	4.60	1.806E+08	1.485E+06	1.83E+00	2.715E+06	2.03E+00	3.010E+06	2.01E+00	2.982E+06	6.94E-01	1.031E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.60	4.80	1.791E+08	1.411E+06	1.71E+00	2.408E+06	1.90E+00	2.680E+06	2.00E+00	2.826E+06	9.50E-01	1.341E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.80	5.20	1.777E+08	2.665E+06	1.55E+00	4.119E+06	1.72E+00	4.595E+06	1.91E+00	5.090E+06	1.24E+00	3.316E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.20	5.60	1.751E+08	2.472E+06	1.36E+00	3.364E+06	1.52E+00	3.758E+06	1.74E+00	4.291E+06	1.52E+00	3.746E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.60	6.00	1.726E+08	2.270E+06	1.21E+00	2.737E+06	1.35E+00	3.057E+06	1.55E+00	3.525E+06	1.59E+00	3.618E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.00	6.40	1.703E+08	2.029E+06	1.08E+00	2.185E+06	1.20E+00	2.431E+06	1.39E+00	2.816E+06	1.53E+00	3.094E+06	1.56E-01	3.157E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.40	6.80	1.683E+08	1.884E+06	9.71E-01	1.829E+06	1.07E+00	2.022E+06	1.24E+00	2.343E+06	1.40E+00	2.642E+06	6.25E-01	1.178E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.80	7.20	1.664E+08	1.785E+06	8.83E-01	1.576E+06	9.70E-01	1.732E+06	1.12E+00	1.998E+06	1.28E+00	2.280E+06	9.68E-01	1.728E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.20	7.60	1.646E+08	1.698E+06	8.10E-01	1.376E+06	8.86E-01	1.505E+06	1.01E+00	1.723E+06	1.16E+00	1.974E+06	1.13E+00	1.924E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.60	8.00	1.629E+08	1.595E+06	7.49E-01	1.194E+06	8.15E-01	1.300E+06	9.26E-01	1.478E+06	1.06E+00	1.691E+06	1.15E+00	1.827E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
8.00	9.00	1.613E+08	3.604E+06	6.65E-01	2.398E+06	7.18E-01	2.589E+06	8.07E-01	2.908E+06	9.16E-01	3.301E+06	1.04E+00	3.734E+06	1.07E-01	3.854E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00
9.00	10.00	1.577E+08	3.155E+06	5.84E-01	1.842E+06	6.20E-01	1.956E+06	6.80E-01	2.146E+06	7.59E-01	2.395E+06	8.74E-01	2.757E+06	8.01E-01	2.527E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
10.00	11.00	1.546E+08	2.188E+06	5.36E-01	1.173E+06	5.59E-01	1.223E+06	5.99E-01	1.311E+06	6.53E-01	1.428E+06	7.41E-01	1.621E+06	7.93E-01	1.736E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
11.00	12.00	1.524E+08	1.971E+06	5.05E-01	9.956E+05	5.20E-01	1.025E+06	5.47E-01	1.079E+06	5.83E-01	1.148E+06	6.43E-01	1.268E+06	7.00E-01	1.379E+06	6.14E-01	1.210E+06	0.00E+00	0.000E+00
12.00	13.00	1.504E+08	1.790E+06	4.86E-01	8.696E+05	4.95E-01	8.871E+05	5.13E-01	9.184E+05	5.37E-01	9.611E+05	5.76E-01	1.031E+06	6.18E-01	1.107E+06	6.30E-01	1.127E+06	0.00E+00	0.000E+00
13.00	14.00	1.486E+08	1.639E+06	4.72E-01	7.735E+05	4.79E-01	7.850E+05	4.91E-01	8.039E+05	5.06E-01	8.287E+05	5.30E-01	8.693E+05	5.57E-01	9.129E+05	5.77E-01	9.448E+05	0.00E+00	0.000E+00
14.00	15.00	1.470E+08	1.509E+06	4.63E-01	6.986E+05	4.68E-01	7.063E+05	4.75E-01	7.171E+05	4.85E-01	7.314E+05	4.99E-01	7.536E+05	5.14E-01	7.756E+05	5.27E-01	7.952E+05	0.00E+00	0.000E+00
15.00	16.00	1.455E+08	1.388E+06	4.57E-01	6.343E+05	4.60E-01	6.389E+05	4.65E-01	6.448E+05	4.70E-01	6.521E+05	4.78E-01	6.626E+05	4.84E-01	6.710E+05	4.88E-01	6.773E+05	0.00E+00	0.000E+00
16.00	18.00	1.441E+08	2.498E+06	4.49E-01	1.122E+06	4.51E-01	1.128E+06	4.54E-01	1.135E+06	4.57E-01	1.141E+06	4.58E-01	1.144E+06	4.56E-01	1.140E+06	4.52E-01	1.129E+06	3.86E-01	9.65E-01
18.00	20.00	1.416E+08	2.198E+06	4.39E-01	9.655E+05	4.41E-01	9.699E+05	4.43E-01	9.739E+05	4.44E-01	9.750E+05	4.41E-01	9.694E+05	4.34E-01	9.548E+05	4.23E-01	9.306E+05	3.83E-01	8.40E-01
20.00	22.00	1.394E+08	2.540E+06	4.32E-01	1.097E+06	4.32E-01	1.098E+06	4.34E-01	1.101E+06	4.34E-01	1.101E+06	4.30E-01	1.091E+06	4.22E-01	1.072E+06	4.09E-01	1.039E+06	3.63E-01	9.22E-01
22.00	24.00	1.369E+08	2.278E+06	4.25E-01	9.678E+05	4.24E-01	9.669E+05	4.25E-01	9.685E+05	4.25E-01	9.677E+05	4.21E-01	9.598E+05	4.13E-01	9.419E+05	4.01E-01	9.137E+05	3.55E-01	8.08E-01
24.00	26.00	1.346E+08	2.062E+06	4.18E-01	8.617E+05	4.18E-01	8.615E+05	4.18E-01	8.614E+05	4.17E-01	8.599E+05	4.14E-01	8.535E+05	4.07E-01	8.385E+05	3.96E-01	8.158E+05	3.53E-01	7.27E-01
26.00	28.00	1.325E+08	1.881E+06	4.11E-01	7.738E+05	4.11E-01	7.736E+05	4.11E-01	7.731E+05	4.10E-01	7.712E+05	4.07E-01	7.658E+05	4.01E-01	7.537E+05	3.91E-01	7.350E+05	3.53E-01	6.63E-01
28.00	30.00	1.306E+08	1.727E+06	4.04E-01	6.982E+05	4.04E-01	6.983E+05	4.04E-01	6.980E+05	4.03E-01	6.962E+05	4.00E-01	6.917E+05	3.95E-01	6.826E+05	3.86E-01	6.673E+05	3.53E-01	6.09E-01
30.00	34.00	1.289E+08	4.012E+06	3.92E-01	1.573E+06	3.92E-01	1.573E+06	3.92E-01	1.573E+06	3.92E-01	1.572E+06	3.90E-01	1.565E+06	3.86E-01	1.549E+06	3.80E-01	1.523E+06	3.53E-01	1.41E-01
34.00	38.00	1.249E+08	3.460E+06	3.77E-01	1.304E+06	3.77E-01	1.305E+06	3.77E-01	1.305E+06	3.77E-01	1.305E+06	3.76E-01	1.300E+06	3.73E-01	1.292E+06	3.69E-01	1.278E+06	3.50E-01	1.21E-01
38.00	42.00	1.214E+08	3.407E+06	3.63E-01	1.236E+06	3.63E-01	1.237E+06	3.63E-01	1.237E+06	3.63E-01	1.238E+06	3.63E-01	1.235E+06	3.61E-01	1.228E+06	3.58E-01	1.219E+06	3.44E-01	1.17E-01
42.00	46.00	1.180E+08	3.362E+06	3.49E-01	1.175E+06	3.50E-01	1.175E+06	3.50E-01	1.176E+06	3.50E-01	1.177E+06	3.50E-01	1.178E+06	3.48E-01	1.171E+06	3.47E-01	1.166E+06	3.37E-01	1.13E-01
46.00	50.00	1.147E+08	2.997E+06	3.36E-01	1.007E+06	3.36E-01	1.007E+06	3.36E-01	1.008E+06	3.37E-01	1.009E+06	3.37E-01	1.010E+06	3.36E-01	1.008E+06	3.35E-01	1.005E+06	3.29E-01	9.85E-01
50.00	55.00	1.117E+08	4.751E+06	3.22E-01	1.530E+06	3.22E-01	1.530E+06	3.22E-01	1.531E+06	3.23E-01	1.533E+06	3.23E-01	1.535E+06	3.23E-01	1.535E+06	3.23E-01	1.534E+0		

1.00	1.20	3.378E+08	3.728E+07	1.89E+00	7.044E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.20	1.30	3.006E+08	1.504E+07	2.04E+00	3.064E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.30	1.40	2.855E+08	1.325E+07	2.12E+00	2.810E+07	9.77E-02	1.295E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.40	1.60	2.723E+08	2.065E+07	2.23E+00	4.594E+07	3.41E-01	7.032E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.60	1.80	2.516E+08	1.540E+07	2.35E+00	3.627E+07	7.63E-01	1.176E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.80	2.00	2.362E+08	1.298E+07	2.46E+00	3.188E+07	1.15E+00	1.489E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.00	2.20	2.232E+08	6.172E+06	2.54E+00	1.565E+07	1.47E+00	9.096E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.20	2.40	2.171E+08	5.486E+06	2.59E+00	1.422E+07	1.74E+00	9.538E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.40	2.60	2.116E+08	4.924E+06	2.63E+00	1.295E+07	1.95E+00	9.609E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.60	2.80	2.067E+08	4.457E+06	2.65E+00	1.181E+07	2.12E+00	9.438E+06	8.86E-02	3.948E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.80	3.00	2.022E+08	4.063E+06	2.65E+00	1.076E+07	2.24E+00	9.116E+06	4.16E-01	1.689E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.00	3.20	1.981E+08	3.296E+06	2.62E+00	8.631E+06	2.34E+00	7.701E+06	7.60E-01	2.505E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.20	3.40	1.948E+08	3.046E+06	2.56E+00	7.803E+06	2.40E+00	7.298E+06	1.07E+00	3.273E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.40	3.60	1.918E+08	2.829E+06	2.48E+00	7.003E+06	2.42E+00	6.840E+06	1.34E+00	3.798E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.60	3.80	1.890E+08	2.637E+06	2.36E+00	6.233E+06	2.40E+00	6.340E+06	1.56E+00	4.123E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.80	4.00	1.863E+08	2.468E+06	2.23E+00	5.503E+06	2.35E+00	5.809E+06	1.74E+00	4.293E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.00	4.20	1.839E+08	1.659E+06	2.09E+00	3.468E+06	2.27E+00	3.764E+06	1.87E+00	3.109E+06	1.28E-01	2.121E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.20	4.40	1.822E+08	1.568E+06	1.96E+00	3.067E+06	2.16E+00	3.379E+06	1.96E+00	3.080E+06	4.10E-01	6.426E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.40	4.60	1.806E+08	1.485E+06	1.83E+00	2.715E+06	2.03E+00	3.010E+06	2.01E+00	2.982E+06	6.94E-01	1.031E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.60	4.80	1.791E+08	1.411E+06	1.71E+00	2.408E+06	1.90E+00	2.680E+06	2.00E+00	2.826E+06	9.50E-01	1.341E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.80	5.20	1.777E+08	2.665E+06	1.55E+00	4.119E+06	1.72E+00	4.595E+06	1.91E+00	5.090E+06	1.24E+00	3.316E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.20	5.60	1.751E+08	2.472E+06	1.36E+00	3.364E+06	1.52E+00	3.758E+06	1.74E+00	4.291E+06	1.52E+00	3.746E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.60	6.00	1.726E+08	2.270E+06	1.21E+00	2.737E+06	1.35E+00	3.057E+06	1.55E+00	3.525E+06	1.59E+00	3.618E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.00	6.40	1.703E+08	2.029E+06	1.08E+00	2.185E+06	1.20E+00	2.431E+06	1.39E+00	2.816E+06	1.53E+00	3.094E+06	1.56E-01	3.157E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.40	6.80	1.683E+08	1.884E+06	9.71E-01	1.829E+06	1.07E+00	2.022E+06	1.24E+00	2.343E+06	1.40E+00	2.642E+06	6.25E-01	1.178E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.80	7.20	1.664E+08	1.785E+06	8.83E-01	1.576E+06	9.70E-01	1.732E+06	1.12E+00	1.998E+06	1.28E+00	2.280E+06	9.68E-01	1.728E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.20	7.60	1.646E+08	1.698E+06	8.10E-01	1.376E+06	8.86E-01	1.505E+06	1.01E+00	1.723E+06	1.16E+00	1.974E+06	1.13E+00	1.924E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.60	8.00	1.629E+08	1.595E+06	7.49E-01	1.194E+06	8.15E-01	1.300E+06	9.26E-01	1.478E+06	1.06E+00	1.691E+06	1.15E+00	1.827E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
8.00	9.00	1.613E+08	3.604E+06	6.65E-01	2.398E+06	7.18E-01	2.589E+06	8.07E-01	2.908E+06	9.16E-01	3.301E+06	1.04E+00	3.734E+06	1.07E-01	3.854E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00
9.00	10.00	1.577E+08	3.155E+06	5.84E-01	1.842E+06	6.20E-01	1.956E+06	6.80E-01	2.146E+06	7.59E-01	2.395E+06	8.74E-01	2.757E+06	8.01E-01	2.527E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
10.00	11.00	1.546E+08	2.188E+06	5.36E-01	1.173E+06	5.59E-01	1.223E+06	5.99E-01	1.311E+06	6.53E-01	1.428E+06	7.41E-01	1.621E+06	7.93E-01	1.736E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00
11.00	12.00	1.524E+08	1.971E+06	5.05E-01	9.956E+05	5.20E-01	1.025E+06	5.47E-01	1.079E+06	5.83E-01	1.148E+06	6.43E-01	1.268E+06	7.00E-01	1.379E+06	6.14E-01	1.210E+06	0.00E+00	0.000E+00
12.00	13.00	1.504E+08	1.790E+06	4.86E-01	8.696E+05	4.95E-01	8.871E+05	5.13E-01	9.184E+05	5.37E-01	9.611E+05	5.76E-01	1.031E+06	6.18E-01	1.107E+06	6.30E-01	1.127E+06	0.00E+00	0.000E+00
13.00	14.00	1.486E+08	1.639E+06	4.72E-01	7.735E+05	4.79E-01	7.850E+05	4.91E-01	8.039E+05	5.06E-01	8.287E+05	5.30E-01	8.693E+05	5.57E-01	9.129E+05	5.77E-01	9.448E+05	0.00E+00	0.000E+00
14.00	15.00	1.470E+08	1.509E+06	4.63E-01	6.986E+05	4.68E-01	7.063E+05	4.75E-01	7.171E+05	4.85E-01	7.314E+05	4.99E-01	7.536E+05	5.14E-01	7.756E+05	5.27E-01	7.952E+05	0.00E+00	0.000E+00
15.00	16.00	1.455E+08	1.388E+06	4.57E-01	6.343E+05	4.60E-01	6.389E+05	4.65E-01	6.448E+05	4.70E-01	6.521E+05	4.78E-01	6.626E+05	4.84E-01	6.710E+05	4.88E-01	6.773E+05	0.00E+00	0.000E+00
16.00	18.00	1.441E+08	2.498E+06	4.49E-01	1.122E+06	4.51E-01	1.128E+06	4.54E-01	1.135E+06	4.57E-01	1.141E+06	4.58E-01	1.144E+06	4.56E-01	1.140E+06	4.52E-01	1.129E+06	3.86E-01	9.65E-01
18.00	20.00	1.416E+08	2.198E+06	4.39E-01	9.655E+05	4.41E-01	9.699E+05	4.43E-01	9.739E+05	4.44E-01	9.750E+05	4.41E-01	9.694E+05	4.34E-01	9.548E+05	4.23E-01	9.306E+05	3.83E-01	8.40E-01
20.00	22.00	1.394E+08	2.540E+06	4.32E-01	1.097E+06	4.32E-01	1.098E+06	4.34E-01	1.101E+06	4.34E-01	1.101E+06	4.30E-01	1.091E+06	4.22E-01	1.072E+06	4.09E-01	1.039E+06	3.63E-01	9.22E-01
22.00	24.00	1.369E+08	2.278E+06	4.25E-01	9.678E+05	4.24E-01	9.669E+05	4.25E-01	9.685E+05	4.25E-01	9.677E+05	4.21E-01	9.598E+05	4.13E-01	9.419E+05	4.01E-01	9.137E+05	3.55E-01	8.08E-01
24.00	26.00	1.346E+08	2.062E+06	4.18E-01	8.617E+05	4.18E-01	8.615E+05	4.18E-01	8.614E+05	4.17E-01	8.599E+05	4.14E-01	8.535E+05	4.07E-01	8.385E+05	3.96E-01	8.158E+05	3.53E-01	7.27E-01
26.00	28.00	1.325E+08	1.881E+06	4.11E-01	7.738E+05	4.11E-01	7.736E+05	4.11E-01	7.731E+05	4.10E-01	7.712E+05	4.07E-01	7.658E+05	4.01E-01	7.537E+05	3.91E-01	7.350E+05	3.53E-01	6.63E-01
28.00	30.00	1.306E+08	1.727E+06	4.04E-01	6.982E+05	4.04E-01	6.983E+05	4.04E-01	6.980E+05	4.03E-01	6.962E+05	4.00E-01	6.917E+05	3.95E-01	6.826E+05	3.86E-01	6.673E+05	3.53E-01	6.09E-01
30.00	34.00	1.289E+08	4.012E+06	3.92E-01	1.573E+06	3.92E-01	1.573E+06	3.92E-01	1.573E+06	3.92E-01	1.572E+06	3.90E-01	1.565E+06	3.86E-01	1.549E+06	3.80E-01	1.523E+06	3.53E-01	1.41E+00
34.00	38.00	1.249E+08	3.460E+06	3.77E-01	1.304E+06	3.77E-01	1.305E+06	3.77E-01	1.305E+06	3.77E-01	1.305E+06	3.76E-01	1.300E+06	3.73E-01	1.292E+06	3.69E-01	1.278E+06	3.50E-01	1.21E+00
38.00	42.00	1.214E+08	3.407E+06	3.63E-01	1.236E+06	3.63E-01	1.237E+06	3.63E-01	1.237E+06	3.63E-01	1.238E+06	3.63E-01	1.235E+06	3.61E-01	1.228E+06	3.58E-01	1.219E+06	3.44E-01	1.17E+00
42.00																			

0.60	0.80	5.754E+08	1.521E+08	2.20E-01	3.339E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.80	1.00	4.233E+08	8.545E+07	4.11E-01	3.516E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.00	1.20	3.378E+08	3.728E+07	6.09E-01	2.272E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.20	1.30	3.006E+08	1.504E+07	7.49E-01	1.126E+07	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.30	1.40	2.855E+08	1.325E+07	8.21E-01	1.088E+07	5.20E-04	6.887E+03	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.40	1.60	2.723E+08	2.065E+07	9.00E-01	1.859E+07	2.13E-02	4.401E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.60	1.80	2.516E+08	1.540E+07	9.81E-01	1.511E+07	1.48E-01	2.286E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
1.80	2.00	2.362E+08	1.298E+07	1.02E+00	1.329E+07	3.42E-01	4.442E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.00	2.20	2.232E+08	6.172E+06	1.04E+00	6.441E+06	5.34E-01	3.298E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.20	2.40	2.171E+08	5.486E+06	1.04E+00	5.730E+06	6.81E-01	3.735E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.40	2.60	2.116E+08	4.924E+06	1.03E+00	5.081E+06	7.81E-01	3.845E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.60	2.80	2.067E+08	4.457E+06	1.01E+00	4.499E+06	8.44E-01	3.761E+06	2.94E-03	1.312E+04	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
2.80	3.00	2.022E+08	4.063E+06	9.80E-01	3.981E+06	8.80E-01	3.573E+06	8.99E-02	3.651E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.00	3.20	1.981E+08	3.296E+06	9.46E-01	3.119E+06	8.94E-01	2.946E+06	2.53E-01	8.324E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.20	3.40	1.948E+08	3.046E+06	9.11E-01	2.776E+06	8.92E-01	2.716E+06	4.09E-01	1.247E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.40	3.60	1.918E+08	2.829E+06	8.77E-01	2.480E+06	8.78E-01	2.485E+06	5.28E-01	1.493E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.60	3.80	1.890E+08	2.637E+06	8.43E-01	2.224E+06	8.59E-01	2.265E+06	6.12E-01	1.613E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
3.80	4.00	1.863E+08	2.468E+06	8.12E-01	2.003E+06	8.36E-01	2.064E+06	6.67E-01	1.645E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.00	4.20	1.839E+08	1.659E+06	7.84E-01	1.301E+06	8.12E-01	1.346E+06	6.99E-01	1.160E+06	1.49E-02	2.478E+04	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.20	4.40	1.822E+08	1.568E+06	7.60E-01	1.192E+06	7.86E-01	1.232E+06	7.15E-01	1.120E+06	1.27E-01	1.987E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.40	4.60	1.806E+08	1.485E+06	7.37E-01	1.095E+06	7.62E-01	1.131E+06	7.19E-01	1.067E+06	2.62E-01	3.897E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.60	4.80	1.791E+08	1.411E+06	7.14E-01	1.007E+06	7.39E-01	1.042E+06	7.15E-01	1.009E+06	3.71E-01	5.235E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
4.80	5.20	1.777E+08	2.665E+06	6.81E-01	1.816E+06	7.07E-01	1.884E+06	7.00E-01	1.867E+06	4.72E-01	1.258E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.20	5.60	1.751E+08	2.472E+06	6.41E-01	1.584E+06	6.67E-01	1.649E+06	6.75E-01	1.669E+06	5.50E-01	1.360E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
5.60	6.00	1.726E+08	2.270E+06	6.04E-01	1.371E+06	6.30E-01	1.430E+06	6.46E-01	1.468E+06	5.78E-01	1.312E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.00	6.40	1.703E+08	2.029E+06	5.72E-01	1.161E+06	5.95E-01	1.208E+06	6.17E-01	1.251E+06	5.82E-01	1.180E+06	3.21E-02	6.521E+04	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.40	6.80	1.683E+08	1.884E+06	5.46E-01	1.029E+06	5.65E-01	1.065E+06	5.87E-01	1.106E+06	5.73E-01	1.079E+06	2.39E-01	4.494E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
6.80	7.20	1.664E+08	1.785E+06	5.23E-01	9.339E+05	5.40E-01	9.641E+05	5.60E-01	1.000E+06	5.58E-01	9.954E+05	3.56E-01	6.352E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.20	7.60	1.646E+08	1.698E+06	5.04E-01	8.554E+05	5.18E-01	8.805E+05	5.37E-01	9.112E+05	5.39E-01	9.161E+05	4.13E-01	7.010E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
7.60	8.00	1.629E+08	1.595E+06	4.87E-01	7.761E+05	5.00E-01	7.970E+05	5.16E-01	8.223E+05	5.21E-01	8.307E+05	4.40E-01	7.017E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
8.00	9.00	1.613E+08	3.604E+06	4.63E-01	1.668E+06	4.74E-01	1.706E+06	4.86E-01	1.752E+06	4.92E-01	1.772E+06	4.48E-01	1.615E+06	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
9.00	10.00	1.577E+08	3.155E+06	4.41E-01	1.390E+06	4.47E-01	1.409E+06	4.53E-01	1.431E+06	4.57E-01	1.443E+06	4.38E-01	1.383E+06	3.12E-01	9.831E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00
10.00	11.00	1.546E+08	2.188E+06	4.28E-01	9.376E+05	4.31E-01	9.425E+05	4.33E-01	9.474E+05	4.32E-01	9.465E+05	4.20E-01	9.183E+05	3.61E-01	7.890E+05	0.00E+00	0.000E+00	0.00E+00	0.000E+00
11.00	12.00	1.524E+08	1.971E+06	4.21E-01	8.305E+05	4.22E-01	8.309E+05	4.21E-01	8.290E+05	4.17E-01	8.214E+05	4.03E-01	7.939E+05	3.66E-01	7.217E+05	2.43E-01	4.781E+05	0.00E+00	0.000E+00
12.00	13.00	1.504E+08	1.790E+06	4.19E-01	7.503E+05	4.18E-01	7.479E+05	4.14E-01	7.408E+05	4.07E-01	7.296E+05	3.91E-01	7.008E+05	3.62E-01	6.479E+05	2.95E-01	5.280E+05	0.00E+00	0.000E+00
13.00	14.00	1.486E+08	1.639E+06	4.18E-01	6.855E+05	4.16E-01	6.822E+05	4.11E-01	6.739E+05	4.03E-01	6.604E+05	3.85E-01	6.309E+05	3.57E-01	5.851E+05	3.10E-01	5.074E+05	0.00E+00	0.000E+00
14.00	15.00	1.470E+08	1.509E+06	4.19E-01	6.320E+05	4.17E-01	6.288E+05	4.10E-01	6.195E+05	4.02E-01	6.060E+05	3.82E-01	5.767E+05	3.55E-01	5.351E+05	3.14E-01	4.747E+05	0.00E+00	0.000E+00
15.00	16.00	1.455E+08	1.388E+06	4.20E-01	5.833E+05	4.18E-01	5.797E+05	4.11E-01	5.706E+05	4.02E-01	5.575E+05	3.82E-01	5.296E+05	3.54E-01	4.915E+05	3.17E-01	4.403E+05	0.00E+00	0.000E+00
16.00	18.00	1.441E+08	2.498E+06	4.20E-01	1.050E+06	4.18E-01	1.044E+06	4.13E-01	1.031E+06	4.04E-01	1.009E+06	3.85E-01	9.608E+05	3.58E-01	8.935E+05	3.23E-01	8.064E+05	1.78E-01	4.44E-01
18.00	20.00	1.416E+08	2.198E+06	4.18E-01	9.192E+05	4.17E-01	9.164E+05	4.13E-01	9.076E+05	4.05E-01	8.912E+05	3.89E-01	8.543E+05	3.64E-01	8.006E+05	3.32E-01	7.305E+05	2.30E-01	5.06E-01
20.00	22.00	1.394E+08	2.540E+06	4.16E-01	1.056E+06	4.14E-01	1.051E+06	4.11E-01	1.044E+06	4.05E-01	1.029E+06	3.91E-01	9.924E+05	3.70E-01	9.409E+05	3.43E-01	8.700E+05	2.54E-01	6.45E-01
22.00	24.00	1.369E+08	2.278E+06	4.12E-01	9.384E+05	4.10E-01	9.340E+05	4.08E-01	9.286E+05	4.03E-01	9.177E+05	3.92E-01	8.923E+05	3.74E-01	8.529E+05	3.51E-01	7.996E+05	2.74E-01	6.24E-01
24.00	26.00	1.346E+08	2.062E+06	4.07E-01	8.396E+05	4.06E-01	8.374E+05	4.04E-01	8.328E+05	4.00E-01	8.244E+05	3.91E-01	8.060E+05	3.76E-01	7.760E+05	3.57E-01	7.360E+05	2.91E-01	5.99E-01
26.00	28.00	1.325E+08	1.881E+06	4.02E-01	7.570E+05	4.02E-01	7.555E+05	4.00E-01	7.521E+05	3.96E-01	7.453E+05	3.89E-01	7.311E+05	3.77E-01	7.085E+05	3.60E-01	6.776E+05	3.04E-01	5.72E-01
28.00	30.00	1.306E+08	1.727E+06	3.97E-01	6.853E+05	3.96E-01	6.843E+05	3.95E-01	6.819E+05	3.92E-01	6.766E+05	3.85E-01	6.658E+05	3.76E-01	6.492E+05	3.62E-01	6.251E+05	3.14E-01	5.42E-01
30.00	34.00	1.289E+08	4.012E+06	3.86E-01	1.548E+06	3.86E-01	1.547E+06	3.85E-01	1.544E+06	3.83E-01	1.537E+06	3.79E-01	1.519E+06	3.72E-01	1.490E+06	3.61E-01	1.449E+06		

Solar protons for $P_{\max}$ (integration restricted to the input energy range). $D(E,..)$ values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

https://www.spennis.oma.be/spennis/htusers/k/kawakita/DEMO1/1778399787/spennis_efp.html 5/8

Solar protons for V_{oc} (integration restricted to the input energy range). $D(E,..)$ values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

			0.00 micron		25.40 micron		76.20 micron		152.40 micron		304.80 micron		508.00 micron		762.00 micron		1524.00 micron	
Energy Int. (MeV)	Int. Fluence (cm ⁻²)	Diff. Fluence (cm ⁻²)	D(E, 0.0)	Eq. Fluence (cm ⁻²)	D(E, 1.0)	Eq. Fluence (cm ⁻²)	D(E, 3.0)	Eq. Fluence (cm ⁻²)	D(E, 6.0)	Eq. Fluence (cm ⁻²)	D(E,12.0)	Eq. Fluence (cm ⁻²)	D(E,20.0)	Eq. Fluence (cm ⁻²)	D(E,30.0)	Eq. Fluence (cm ⁻²)	D(E,60.0)	Eq. Fluence (cm ⁻²)
0.10	0.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00
0.20	0.30	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
0.30	0.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
0.40	0.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
0.60	0.80	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
0.80	1.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
1.00	1.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
1.20	1.30	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
1.30	1.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
1.40	1.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
1.60	1.80	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
1.80	2.00	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
2.00	2.20	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
2.20	2.40	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
2.40	2.60	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00
2.60	2.80	0.000E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00	0.00E+00	0.00E+00	0.000E+00	0.00E+00	0.000E+00	0.00E+00

simplify the output information; the accuracy of the results should not be affected by this simplification.

[illegible]

Solar protons for I_{sc} (integration restricted to the input energy range). $D(E,..)$ values are coarse averages introduced into the table to simplify the output information; the accuracy of the results should not be affected by this simplification.

https://www.spenvis.oma.be/spenvis/htusers/k/kawakita/DEMO1/1778399787/spenvis_efp.html 7/8

[illegible]